

CS 486/686

Introduction to Machine Learning

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Lecture 16

RN 19.1–19.2 · PM 7.1–7.2

[Search](#)

›

[Uncertainty](#)

›

[Decisions](#)

›

[Learning](#)

Learning goals

- Identify reasons for building an agent that can **learn**.
- Describe different **types of learning** (supervised, unsupervised, RL).
- Define **supervised learning, classification, and regression**.
- Define **bias, variance**, and describe the trade-off.
- Use **cross-validation** to prevent overfitting.

Why an agent that learns?



Medical
diagnosis



Spam filtering



Face
recognition



Speech



Handwriting

Learning = the ability to improve on future tasks based on experience.

- We can't anticipate every situation.
- We can't anticipate every change over time.
- For many tasks, we have no idea how to *program* a solution — only how to *show examples*.

The learning architecture

Every learning system is defined by four ingredients:

Task

The behaviour we want to improve (classify, predict, decide).

Data

Experience used to improve (examples, demonstrations, rewards).

Background knowledge / bias

What we assume up front (model class, smoothness, priors).

Measure of improvement

How we score progress (accuracy, speed, new abilities).

Three kinds of learning

Supervised



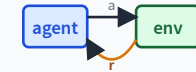
Given input + target features, predict targets for new inputs.
Today's focus.

Unsupervised



No targets — find structure: clustering, dimensionality reduction.
L17.

Reinforcement



Learn from rewards and punishments — what to *do*.
L15.

Supervised or unsupervised?

Q1. We have a user's credit-card transactions and want to flag any that look *different* from the rest. We have no fraud labels.

- A. Supervised
- B. **Unsupervised**

B — Unsupervised. No target labels to predict.

Q2. We have historical weather labels (sunny/cloudy/rain/snow) for a date, and we want to predict the same date next year.

- A. **Supervised**
- B. Unsupervised

A — Supervised. Each example has a target (the weather label).

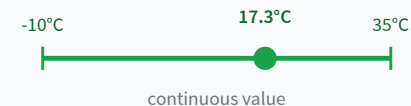
Two flavors of supervised learning

Classification



Target features are **discrete** — e.g. dog vs cat.

Regression



Target features are **continuous** — e.g. tomorrow's temperature.

Quick check: classification or regression?

Q3. Historical weather labels (sunny / cloudy / rain / snow); predict next year's label for a given date.

A. **Classification**

B. Regression

A — Classification. The target takes a discrete value from a fixed set.

Q4. Historical house prices over time; predict next month's price for a particular house.

A. Classification

B. **Regression**

B — Regression. The target is a continuous real number.

Supervised learning = function approximation

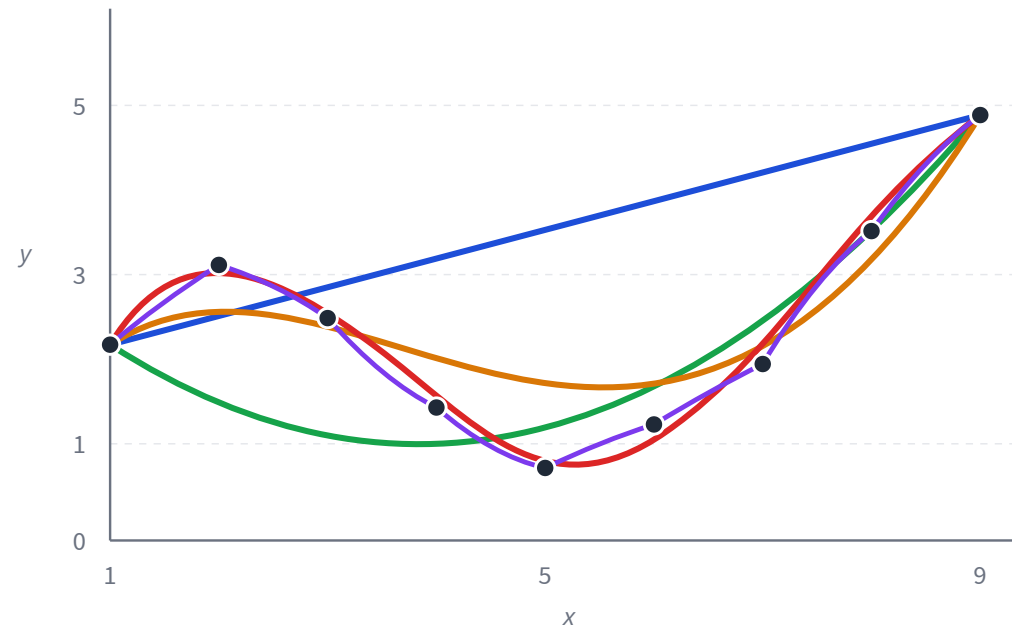
Given training examples $(x_i, f(x_i))$, find a **hypothesis** h that approximates the unknown true function $f : X \rightarrow Y$.

A **hypothesis space** is the set of functions we'll consider (e.g. all polynomials of degree $\leq k$).

Learning = a search problem in the hypothesis space — usually some form of local search (gradient descent, etc.).

Example: fitting a polynomial

Same 9 data points, five candidate polynomial models. Watch how the fit changes with complexity.



— degree 1 (line) — degree 2 — degree 3 — degree 5 — degree 9 (overfits)

Which curve is "right"?

- If the data has **outliers** or is noisy, simpler curves are more trustworthy.
- If the data really does have wild swings (stock prices, etc.), the complex curve is closer to truth.
- **There is no perfect answer** — each curve is justified *by some assumption*.

No-free-lunch

To learn something useful, we *must* make assumptions. Those assumptions are the **inductive bias** of the learner.

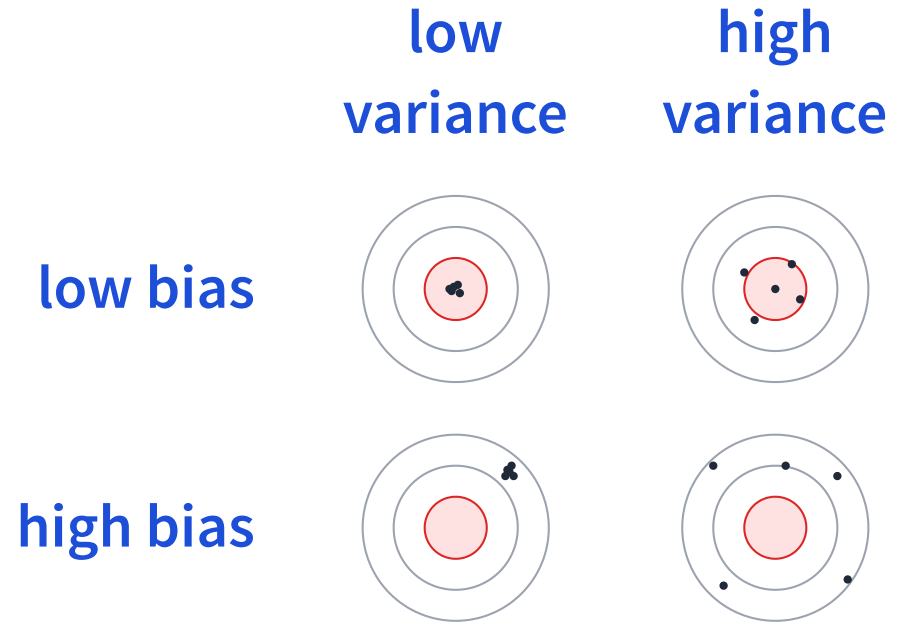
Bias and variance

Bias

With infinite data, how well can the family fit f ?
High bias = too simple, **underfits**.

Variance

How much does the learned hypothesis change with different training sets? High variance = too flexible, **overfits**.



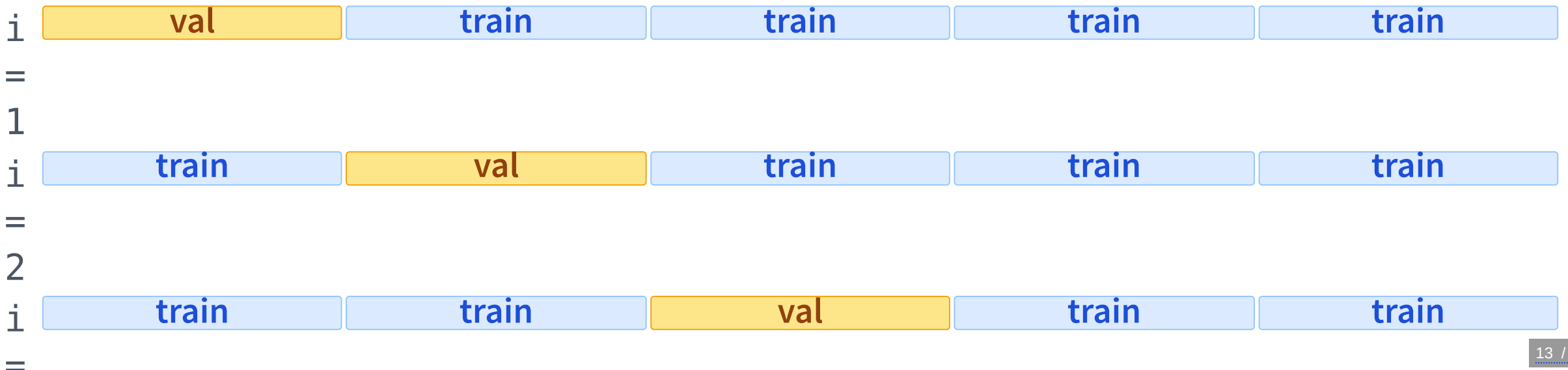
Dots = predictions from re-trained models.
Bullseye = truth.

K-fold cross-validation

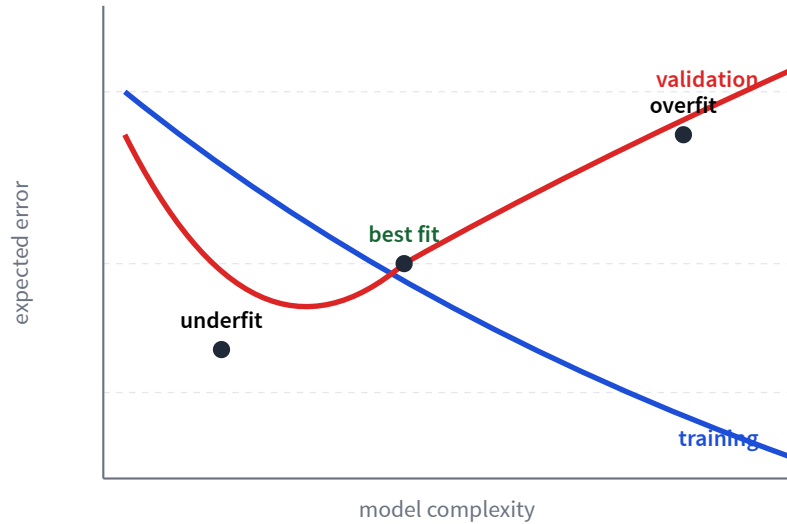
How to pick a hypothesis that generalizes? Use a slice of training data as a **surrogate test set**.

k-fold-CV(model, dataset, K)

1. Split training data into K equally sized folds.
2. For $i = 1, \dots, K$: train on the other $K - 1$ folds; evaluate on fold i (held-out validation).
3. Average error over the K runs.



Overfitting: training vs validation error



Q5. Which is more likely to overfit?

- A. Straight line (simple)
- B. 4th-degree polynomial (complex)
- C. I don't know

B — complex. More degrees of freedom = more wiggle = it can chase the noise.

True error vs empirical error

Setup: domain X , label space Y , unknown true function $f : X \rightarrow Y$. Examples drawn from distribution D over X .

True error (what we want)

$$L_{D,f}(h) = \Pr_{x \sim D} [h(x) \neq f(x)]$$

The probability of being wrong on a fresh random example. We **can't** compute this — we don't know D or f .

Empirical error (what we measure)

$$L_S(h) = \frac{|\{i \in [m] : h(x_i) \neq y_i\}|}{m}$$

Empirical risk minimization (ERM)

Given a hypothesis class $\mathcal{H}$ and a loss ℓ , pick the hypothesis with smallest training loss:

$$\hat{h} = \arg \min_{h \in \mathcal{H}} \frac{1}{m} \sum_{i=1}^m \ell(h(x_i), y_i)$$

Three factors determine how well ERM generalizes:

Data size m

Larger is better: more examples $\Rightarrow$ closer empirical $\rightarrow$ true.

Hypothesis space $\mathcal{H}$

Trade-off: small $\mathcal{H}$ = high bias; large $\mathcal{H}$ = high variance.

Loss function ℓ

Encodes what kind of error you care about (0/1, squared, hinge, ...).

Learning goals (recap)

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Next: unsupervised learning

Today: labels guided every hypothesis we picked. Without labels, what *structure* can we still find in data?

L17: clustering, k-means, autoencoders.